WorkFlow — Team Task Distribution

Project: Web-Based Project and Task Management System

Difficulty Level	Team Member	Main Responsibility	Detailed Tasks
1. Easiest	Ahmed Mohamed	Dashboard, Announcements & Shared UI	**1. Shared UI and Layout** • Build the shared application layout using React, TypeScript, and Tailwind CSS. • Create reusable UI components and consistent styling. • Implement shared navigation and layout elements. • Ensure that the main interfaces are responsive. **2. Dashboard** • Build the Dashboard interface. • Display relevant summaries of projects, tasks, and leave requests. • Organize information clearly for each role. **3. Announcements** • Build the announcements page. • Display company announcements. • Implement announcement creation and editing interfaces for authorized users. **Additional Responsibility** • Assist with integrating and testing simple APIs for announcements or Dashboard

			data in coordination with the Backend member.
2. Easy	Arsany Amir	Employees & Leave Requests	**1. Employee Management** • Build the employee list and employee information interfaces. • Implement interfaces for viewing and editing employee information according to permissions. • Connect the interfaces to the relevant Backend APIs. **2. Leave Requests** • Build the interface for submitting leave requests. • Allow employees to view their request statuses. • Build the interface for managers to review leave requests. • Display pending, approved, and rejected requests. **Additional Responsibility** • Test leave request workflows, including submission, approval, rejection, and validation. • Coordinate with the Backend member to ensure that the required operations work correctly.
3. Medium	Fady Ibrahim	Projects & Tasks	**1. Project Management** • Build the projects list and project details interfaces. • Implement interfaces for creating and editing projects according to permissions. • Display relevant project information and progress. **2. Task Management** • Build the tasks list and task details interfaces. • Implement task creation, editing,

			and assignment to employees. • Support the following task statuses: To Do, In Progress, Review, and Completed. • Display task information and progress clearly. **3. Backend Integration** • Connect project and task interfaces to the relevant APIs. • Handle loading states, errors, and successful operations. **Additional Responsibility** • Test project and task workflows, including task assignments, status changes, and data updates.
4. Hard	Ziad Asem	Authentication & Authorization	**1. Frontend and Backend Integration** • Coordinate with all team members to connect frontend modules to the Backend APIs. • Check that data is transferred and displayed correctly. • Help resolve integration errors between modules. **2. System Testing** • Test projects, tasks, employees, leave requests, announcements, and Dashboard workflows. • Identify bugs and document steps to reproduce them. • Verify input validation, error handling, and expected system behavior. • Retest features after bugs are fixed. **3. Role-Based Access Testing** • Test the Employee, Manager, and Admin permissions. • Verify that each role can perform

			only its permitted actions. • Test attempts to access unauthorized pages, data, and API endpoints. • Coordinate with the Backend member to resolve permission-related issues. **Additional Responsibility** • Help prepare the system for final integration and assist with checking the Docker setup if Docker is used.
5. Hardest	Ziad Mohamed	Backend & Database	**1. Backend Setup** • Set up the Node.js and Express backend structure. • Organize routes, controllers, middleware, and other necessary backend modules. **2. Database Design** • Design MongoDB collections for users, projects, tasks, leave requests, and announcements. • Define the required relationships and references between records. • Support the required create, read, update, and delete operations. **3. API Development** • Build APIs for project and task management. • Build APIs for employee management and leave requests. • Build APIs for announcements and Dashboard data where needed. • Validate incoming data and handle errors consistently. **4. Authentication and**

			Authorization • Implement or integrate a secure method of identifying users, even though custom Login and Sign Up pages will not be developed. • Implement Role-Based Access Control (RBAC) for Employee, Manager, and Admin. • Enforce permissions on the server for every protected API request. • Ensure that users cannot grant themselves a higher role or access unauthorized records. • Apply the principle of least privilege and deny access by default. **5. Integration and Documentation** • Document API endpoints and their expected request and response formats. • Help frontend members integrate their interfaces with the Backend. • Troubleshoot database, API, and access-control issues. • Coordinate with the Integration and Testing member to resolve system-wide problems. **Additional Responsibility** • Coordinate API requirements and help integrate Dashboard and announcements features with the frontend.

Shared Responsibilities — All Team Members

Area	Requirements
Technology Stack	Use React, TypeScript, and Tailwind CSS for the frontend; Node.js and Express for the backend; MongoDB for the database; Git and GitHub for version control and collaboration. Docker is planned for containerization.
Code Collaboration	Use Git and GitHub to share code, review changes, and resolve conflicts. Agree on branch and pull request practices before development.
Integration	Coordinate API formats, component interfaces, and data structures before connecting individual modules.
Testing	Test assigned features and work together to test the complete system, including role-based permissions and error cases.
Responsive Design	Ensure the main interfaces are usable on desktop, tablet, and mobile screen sizes.
Documentation	Document assigned modules, relevant APIs, setup instructions, and known limitations.
Knowledge Sharing	Every member should understand the overall system and review how their module connects to the other modules.
Final Delivery	Integrate the system, fix important issues, complete the documentation, and prepare the final presentation by the agreed deadline of 4 December 2026.